

Pattern Making Studies

A Parametric Approach to Flat Pattern Drafting with R

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Preface

About This Book

This book documents my journey learning pattern making through a parametric, code-based approach. Rather than traditional paper drafting, I translate pattern making instructions into R code, creating:

- Reproducible patterns that adjust to any measurement set
- Fully annotated diagrams with all construction points visible
- A searchable reference of drafting techniques
- A record of errors, corrections, and learning insights

Why Parametric Pattern Making?

Traditional pattern making relies on manual drafting for each size and design. This approach:

1. **Eliminates manual grading:** Change measurements, regenerate the pattern
2. **Preserves design logic:** Every calculation is documented in code
3. **Enables experimentation:** Test design changes instantly
4. **Creates a knowledge base:** Build upon previous work systematically

Tools Used

- **R and R Markdown/Bookdown:** Documentation and computation
- **SeamlyMe:** Body measurement management
- **FreeSewing Notation:** Standard pattern marking symbols
- **Git/GitHub:** Version control and sharing

How to Use This Book

The book follows a structured learning path:

1. **Chapter 0:** Notation standards and conventions
2. **Chapter 1:** Measurement systems and file formats
3. **Chapters 2a-2f:** Basic blocks for different garments
4. **Chapter 3:** Design variations from basic blocks
5. **Chapter 4:** Real projects with iterations and corrections
6. **Chapter 5:** Reference tables and formulas

Prerequisites

- R (≥ 4.0) and RStudio
- LaTeX distribution (for PDF output)
- Basic understanding of garment construction
- Optional: SeamlyMe for measurement management

Acknowledgments

- FreeSewing.org for the notation system
- Traditional pattern making textbooks
- Open source community

1. Notation Standards and Conventions

This chapter defines all notation, terminology, and standards used throughout this pattern making documentation. Following these conventions ensures consistency across all pattern blocks and variations.

1.1. Measurement System

All measurements in this book use the **metric system** exclusively.

Unit	Abbreviation	Precision	Usage
Centimeter	cm	0.1	Body measurements, pattern dimensions, ease
Millimeter	mm	0.5	Notches, seam allowances, small details
Meter	m	0.01	Fabric length estimation only

Critical rule: Never mix measurement systems. All formulas assume centimeter inputs and return centimeter outputs.

1.2. Coordinate System

All patterns are drafted on a 2D Cartesian coordinate system following textile industry conventions:

- **Origin (0,0):** Bottom-left corner of the foundation rectangle (point D or C depending on orientation)
 - **X-axis:** Horizontal, positive values extend to the right (pattern width)
 - **Y-axis:** Vertical, positive values extend upward (pattern length/height)
- Y-axis (Height) ^ | | A(0, height)

height) | | FRONT BACK | | D(0, 0) C(width, 0) | + > X-axis (Width)

Origin (0,0)

1.3. Point Naming Conventions

Construction points follow traditional pattern drafting conventions with letter designations:

1.3.1. Primary Structure Points (Foundation Rectangle)

Point	Description	Typical Position
A	Top-left corner	(0, height)
B	Top-right corner	(width, height)
C	Bottom-right corner	(width, 0)
D	Bottom-left corner	(0, 0)

1.3.2. Construction Division Points

Point	Description	Derivation
E, F	Side seam offset	A + offset, D + offset
G, H	Center/Armscye division	Midpoint of EB, FC
I, J, K, L	Front piece divisions	Divide AG into 4 equal parts

1.3.3. Pattern Feature Points

Category	Points	Represents
Neckline	N, Decote points	Front and back neckline curves
Shoulder	M, P, I, L	Shoulder seam endpoints
Armscye	Q, R, S, T, U, V, X, Y, Z	Sleeve cap curve control points
Darts	Various	Dart apex and legs
Waist/Hip	W, Hip points	Waistline and hip curve points

1.4. Line Types and Meanings

Based on the FreeSewing Notation System, adapted for R/ggplot2:

1.4.1. Structural Lines

Line Type	R Linetype	Color	Purpose
Seam line	<code>solid</code>	Black (<code>#212121</code>)	Actual sewing line
Seam allowance	<code>dashed</code>	Grey (<code>#757575</code>)	Fabric cutting line
Construction line	<code>dotted</code>	Light grey	Temporary guide lines
Grainline	<code>dotted + arrow</code>	Green (<code>#388E3C</code>)	Fabric grain direction
Cut-on-fold	<code>twodash + arrows</code>	Red (<code>#D32F2F</code>)	Place on fabric fold
Center line	<code>dashed</code>	Grey	Center front/back

1.4.2. Annotation Lines

Line Type	R Linetype	Color	Purpose
Dimension	<code>dotted thin</code>	Blue (<code>#1565C0</code>)	Measurement callouts
Note	<code>dashed thin</code>	Dark grey	Construction notes
Mark	<code>solid thin</code>	Various	Placement indicators
Contrast	<code>solid colored</code>	Cyan (<code>#00BCD4</code>)	Highlight specific areas
Help	<code>dotted thin</code>	Light grey	Assembly guides

1.5. Notches and Markings

1.5.1. Notch Types (FreeSewing Standard)

- **Front notch** (): Circle with dot center — indicates front of garment
- **Back notch** (): Circle with cross — indicates back of garment

1.5.2. Construction Marks

Symbol	R Implementation	Meaning
	<code>add_notch(type="front")</code>	Front orientation
	<code>add_notch(type="back")</code>	Back orientation
	<code>add_button()</code>	Button placement
	<code>add_buttonhole()</code>	Buttonhole placement
	<code>add_bartack()</code>	Reinforcement stitch
	<code>add_grainline()</code>	Grain direction

1.6. Pattern Piece Information Block

Every pattern piece must include:

- 1. **Piece identifier:** Number and name (e.g., "1 — Front Bodice")
- 2. **Size:** Measurement set identifier
- 3. **Cutting instructions:** Quantity and fabric type
- 4. **Date:** Drafting date (ISO format: YYYY-MM-DD)
- 5. **Scale verification:** 5cm × 5cm test square

1.7. Color Standards

1.7.1. Fabric Type Colors

Color Name	Hex Code	R Variable	Usage
Primary fabric	#212121	fabric_primary	Main garment fabric
Lining	#1976D2	fabric_lining	Lining pieces
Interfacing	#F57C00	fabric_interfacing	Interfacing/fusing

1.7.2. Annotation Colors

Purpose	Hex Code	R Variable
Seam allowance	#757575	seam_allowance
Grainline	#388E3C	grainline
Cut on fold	#D32F2F	cut_on_fold
Dimensions	#1565C0	dimensions
Front notch	#212121	notch_default
Back notch	#D32F2F	notch_back

1.8. Grid System

All pattern plots include a reference grid for visual measurement:

- **Major grid lines:** Every 5 cm, solid light grey (#E0E0E0), linewidth 0.3
- **Minor grid lines:** Every 1 cm, very light grey (#F5F5F5), linewidth 0.1
- **Axis labels:** Centered on each major grid line

The grid serves as: - Quick visual reference for dimensions - Scale verification tool - Alignment guide for pattern pieces

1.9. File Naming Conventions

{garment}{block}{version}_{size}.{ext}

Examples: bodice_front_female_v1.0_size48.pdf sleeve_basic_v2.1_universal.csv skirt_pencil_variation_v1.0_size42_A0.pdf

1.9.1. Version Numbering

- **Major** (X.0.0): Complete redraft or design change
- **Minor** (0.X.0): Significant adjustments to fit
- **Patch** (0.0.X): Small corrections, annotation fixes

1.10. Printing Standards

1.10.1. Scale Verification

Always print and measure the 5cm test square before cutting. If the square doesn't measure exactly 5cm × 5cm: 1. Check printer settings for "Actual size" (not "Fit to page") 2. Verify PDF scaling is set to 100% 3. Re-print if scale is off by more than 1mm

1.10.2. Paper Sizes

Format	Dimensions (cm)	Use Case
A4	21.0 × 29.7	Home printing (tiled)
A3	29.7 × 42.0	Small patterns
A0	84.1 × 118.9	Plotter, full patterns

1.10.3. Tile Assembly

For tiled home printing: 1. Trim along registration marks 2. Align matching marks between tiles 3. Tape from center outward 4. Verify scale after assembly

1.11. Measurement File Format

1.11.1. SeamlyMe CSV Structure

```
code,reference,description,value,formula
height,A01,Height: Total,154,154
bust_circ,G04,Bust circumference,113,113
```

1.11.2. Ease Parameters CSV Structure

```
parameter,value,unit,description
bust_ease,6,cm,Overall bust ease
shoulder_slope_front,4,cm,Front shoulder drop
```

1.12. R Implementation Notes

1.12.1. Function Naming

`calc_*`: Pure mathematical calculations (distance, angle)

`create_*`: Generate new objects (curves, plots)

`add_*`: Modify existing plots with annotations

`read_*` / `write_*`: File I/O operations

`validate_*`: Check data integrity

1.12.2. Coordinate Storage

Points stored as named vectors: `c(x, y)`

Point collections as named lists: `list(A = c(0, 44), B = c(48, 44))`

Plotted points as data frames: `data.frame(name, x, y)`

1.12.3. Cache Strategy

All plots cached in `images/cache/`

Cache invalidation based on parameter hash

Manual cache clear: `delete images/cache/*.pdf`

1.13. References

FreeSewing Notation Guide: <https://freesewing.eu/docs/about/notation/>

ISO 8559: Garment construction and anthropometric surveys

ABNT NBR 13377: Brazilian standard for garment sizes (where applicable)

2. Measurement Systems and Integration

This chapter covers how body measurements are captured, stored, and integrated into the parametric pattern drafting system. We use SeamlyMe as the primary measurement tool, but the system accepts any CSV following our format.

2.1. Measurement Philosophy

2.1.1. Body Measurements vs Pattern Measurements

Body measurements are taken directly from the body (or a dress form). They represent the actual dimensions of the person who will wear the garment.

Pattern measurements are body measurements plus ease. They represent the dimensions of the finished garment.

Pattern Measurement = Body Measurement + Ease

2.1.2. Types of Ease

Ease Type	Purpose	Example
Wearing ease	Minimum needed for movement and comfort	4-6 cm at bust
Design ease	Style choice affecting silhouette	Additional 10 cm for oversized look
Negative ease	Garment smaller than body (stretch fabrics)	-2 cm at bust for fitted knit

2.2. SeamlyMe Integration

SeamlyMe is open-source software for managing body measurements. It exports measurements in a CSV format we can read directly.

2.2.1. Export Process from SeamlyMe

1. Open your measurement file (.vit) in SeamlyMe
2. Go to **Measurements** → **Export to CSV**
3. Select all measurements needed for the pattern
4. Save the CSV to data/measurements/

2.2.2. SeamlyMe CSV Format

```
code,reference,description,value,formula
height,A01,Height: Total,154,154
bust_circ,G04,Bust circumference,113,113
neck_front_to_waist_f,H01,Neck Front to Waist Front,58,58
```

Columns explained:

code: Internal identifier used in our R code (e.g., bust_circ)

reference: SeamlyMe reference code (e.g., G04)

description: Human-readable description

value: The actual measurement in centimeters

formula: Can be a fixed value or a formula referencing other measurements

Required Measurements by Block Type

Female Bodice Block Requirements

3. Required measurements for female bodice

```
female_bodice_required <- c(  
  "height",  
  "bust_circ",  
  "neck_front_to_waist_f",  
  "back_waist_length",  
  "shoulder_length",  
  "armscye_circ",  
  "neck_width",  
  "across_chest_f",  
  "bustpoint_to_bustpoint",  
  "bustpoint_to_neck_side"  
)
```

4. Create reference table

```
reqs_df <- data.frame(
  Code = female_bodice_required,
  Description = c(
    "Total height",
    "Bust circumference",
    "Front waist length (neck to waist)",
    "Back waist length",
    "Shoulder length",
    "Armscye circumference",
    "Neck width",
    "Across chest (front)",
    "Bust point to bust point",
    "Bust point to neck side"
  ),
  Category = c("Reference", "Circumference", "Length", "Length",
    "Length", "Circumference", "Width", "Width",
    "Width", "Length")
)

knitr::kable(reqs_df,
  caption = "Required measurements for female bodice block",
  booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Male Bodice Block Requirements

```
male_bodice_required <- c(
  "height",
  "chest_circ",
  "neck_front_to_waist_f",
  "back_waist_length",
  "shoulder_length",
  "armscye_circ",
  "neck_width",
  "across_back",
  "across_chest_m"
)

reqs_male_df <- data.frame(
  Code = male_bodice_required,
  Description = c(
    "Total height",
    "Chest circumference",
    "Front waist length",
    "Back waist length",
    "Shoulder length",
    "Armscye circumference",
    "Neck width",

```

```

    "Across back width",
    "Across chest (male)"
  ),
  Category = c("Reference", "Circumference", "Length", "Length",
               "Length", "Circumference", "Width", "Width", "Width")
)

knitr::kable(reqs_male_df,
             caption = "Required measurements for male bodice block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))

```

Skirt Block Requirements

Table 4.1: Required measurements for skirt block

Code	Description	Category
waist_circ	Waist circumference	Circumference
hip_circ	Hip circumference	Circumference
waist_to_hip	Waist to hip distance	Length
waist_to_knee	Waist to knee distance	Length
height	Total height	Reference

Sleeve Block Requirements

Table 4.2: Required measurements for sleeve block

Code	Description	Category
armscye_circ	Armscye circumference	Circumference
arm_upper_circ	Upper arm circumference	Circumference
arm_elbow_circ	Elbow circumference	Circumference
arm_wrist_circ	Wrist circumference	Circumference
arm_shoulder_tip_to_wrist	Shoulder to wrist length	Length
arm_shoulder_tip_to_elbow	Shoulder to elbow length	Length
arm_armpit_to_wrist	Armpit to wrist (inside)	Length

Pants Block Requirements

Table 4.3: Required measurements for pants block

Code	Description	Category
waist_circ	Waist circumference	Circumference
hip_circ	Hip circumference	Circumference
waist_to_hip	Waist to hip distance	Length
crotch_depth	Crotch depth (sitting)	Length
inseam	Inseam length	Length
outseam	Outseam length	Length
thigh_circ	Thigh circumference	Circumference
knee_circ	Knee circumference	Circumference
ankle_circ	Ankle circumference	Circumference

Loading Measurements in R

The `read_measurements()` Function

Key measurements for female size 48:

```
## height : 154.0 cm
## bust_circ : 113.0 cm
## neck_front_to_waist_f : 58.0 cm
## shoulder_length : 10.5 cm
```

Creating a Measurement Template

If you don't have SeamlyMe, you can create a template CSV and fill in values manually:

Measurement Validation

Checking Required Measurements

```
## objeto 'female_bodice_required' não encontrado
```

```
## Consistency checks:
```

```
## waist_vs_bust
## hip_vs_waist
## arm_segments
```

Ease Parameters

Ease parameters are stored separately from body measurements.

This separation allows:

Changing body measurements without affecting design decisions

Experimenting with different fits on the same body

Sharing ease settings across measurement sets

Ease Parameters File Format

Ease by Garment Type and Fit Garment Close Fit Semi-Fit Loose Fit Oversized Bodice bust 4-6 cm 6-8 cm 8-12 cm 12-20 cm
Bodice waist 2-4 cm 4-6 cm 6-10 cm 10-16 cm Skirt hip 2-4 cm 4-6 cm 6-8 cm 8-12 cm Pants hip 2-4 cm 4-6 cm 6-8 cm — Sleeve
biceps 2-4 cm 4-6 cm 6-8 cm 8-12 cm Standard Size Reference Tables Female Standard Measurements (ABNT NBR 13377)

Table 4.4: Standard female measurements (cm) — Reference only

Size	Bust	Waist	Hip	Neck_to_Waist_F	Back_Waist
38	84	64	92	44	38
40	88	68	96	45	39
42	92	72	100	46	40
44	96	76	104	47	41
46	102	82	110	48	42
48	108	88	116	50	43
50	114	94	122	52	44

Male Standard Measurements

Table 4.5: Standard male measurements (cm) — Reference only

Size	Chest	Waist	Hip	Back_Waist
46	92	78	94	44
48	96	82	98	45
50	100	86	102	46
52	104	90	106	47
54	108	96	110	48
56	112	102	114	49

Practical Measurement Tips Taking Body Measurements

Use a flexible tape measure - not a metal one

Measure over undergarments - or the clothing layer the garment will be worn over

Keep the tape parallel to the floor - especially for circumferences

Don't pull too tight - the tape should be snug but not compressing

Mark landmarks - use elastic bands to mark waist, hip, and bust levels

Take multiple measurements - average three readings for accuracy

Record posture - note if the subject has forward shoulders, sway back, etc.

Measurement Landmarks Landmark How to Find
Waist level Side bend — where the body creases
Hip level Widest part of hips, usually 18-22 cm below waist
Bust point Apex of bust (nipple level)
Neck base Where neck meets shoulder
Shoulder tip Acromion bone (bony point at shoulder edge)
Elbow Olecranon process (bony point of elbow)
Wrist Styloid process (bony bump at wrist)

Chapter Summary

All measurements in centimeters, stored in SeamlyMe CSV format

Measurements are validated for presence and consistency before drafting

Ease parameters are stored separately for flexibility

Standard size tables provided for reference only - use actual body measurements when possible

Each block type has specific measurement requirements documented above